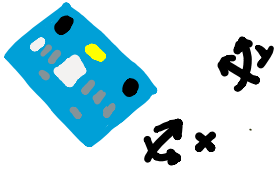


Mpu 6050 6-axis IMU



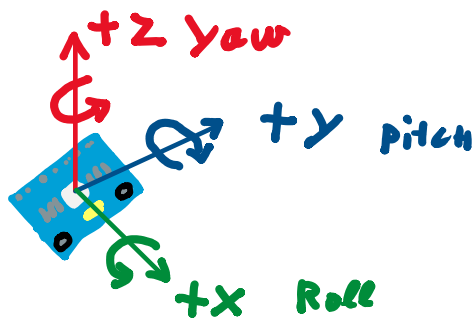
- 3 - Axis Gyroscope
- 3 - Axis Accelerometer

- I2C
- SPI

Gyroscope

Meets Rotational Motion.

- Angular velocity (Rotation speed) in degrees per seconds $[^{\circ}/s]$.
 - X-axis [Roll]
 - Y-axis [Pitch]
 - Z-axis [Yaw]



Accelerometer

Accelerometer

Measuring Linear acceleration

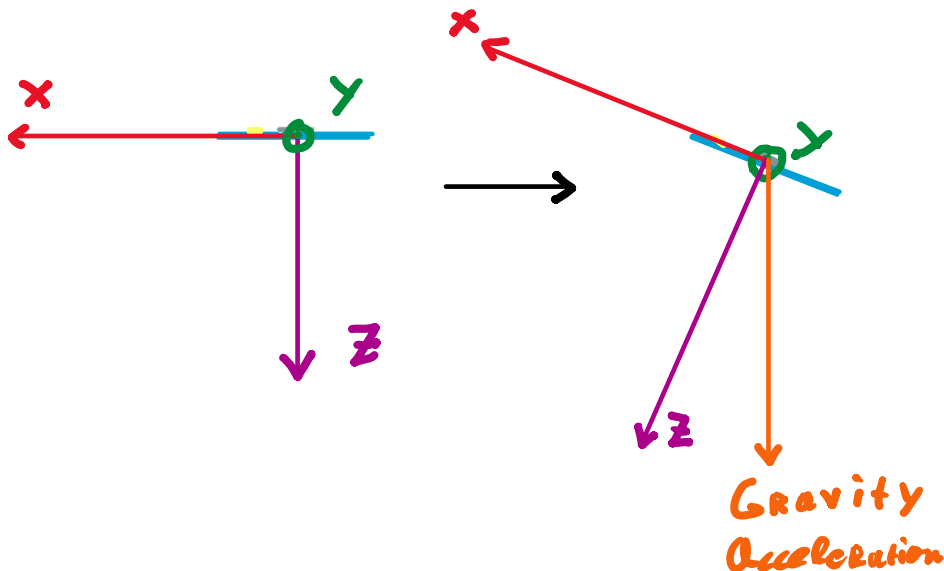
- How fast Velocity is changing in a straight line.

Acceleration

- X - axis [Left/Right acceleration]
- Y - axis [Forward/Backward acceleration]
- Z - axis [Up/down acceleration]

in m/s^2 or g-force

$$[1g = 9.8 m/s^2]$$



- Measuring angles using Gyroscope

Roll Pitch Yaw

- Measuring angles using gyroscope

Stel 5 $^{\circ}/s$ met sample $T_s = 0,1s$

$$5 \text{ } ^{\circ}/s \cdot 0,1s = 5 \cdot 0,1 \frac{\cancel{s}}{\cancel{s}} = 0,5^{\circ} \text{ graden}$$

Read Gyro at 10 Hz $\Delta t = 0,1 \text{ Second}$

timeStep	Measured Rate	Roll angle
0	0	$0 + \Delta t \cdot \text{Rate}$
1	3 $^{\circ}/s$	0,3 deg
2	3 $^{\circ}/s$	0,6 deg
3	3 $^{\circ}/s$	0,9 deg
4	3 $^{\circ}/s$	1,2 deg
5	3 $^{\circ}/s$	1,5 deg
6	3 $^{\circ}/s$	1,8 deg
7	3 $^{\circ}/s$	2,1 deg
8	3 $^{\circ}/s$	2,4 deg
9	3 $^{\circ}/s$	2,7 deg
10	3 $^{\circ}/s$	3 deg

• Called Dead Reckoning

into continuous angular Rate

Integrating angular Rate



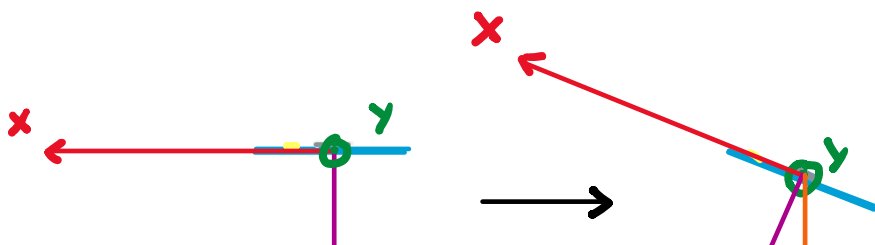
Nadelen

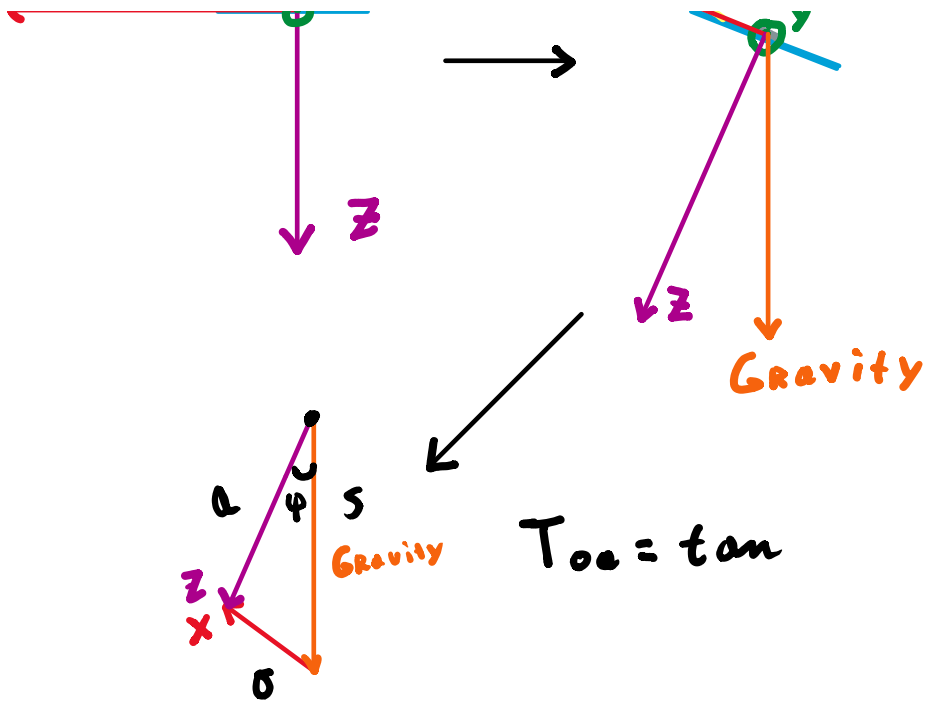
- Drift (Accumulates ERROR)
- Sensor Bias/offset
- No Absolute Reference
- Integration ERROR

- Measuring angles using Accelerometer

Gravity is always 1g (Down)

Pitch - angle using X and Z

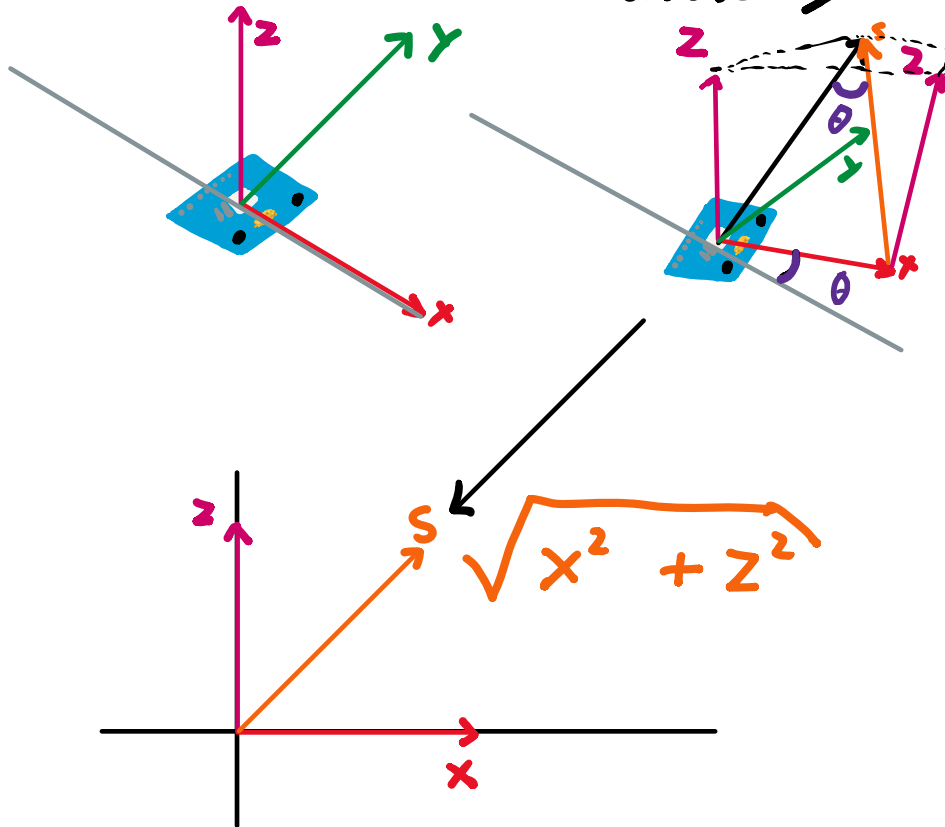


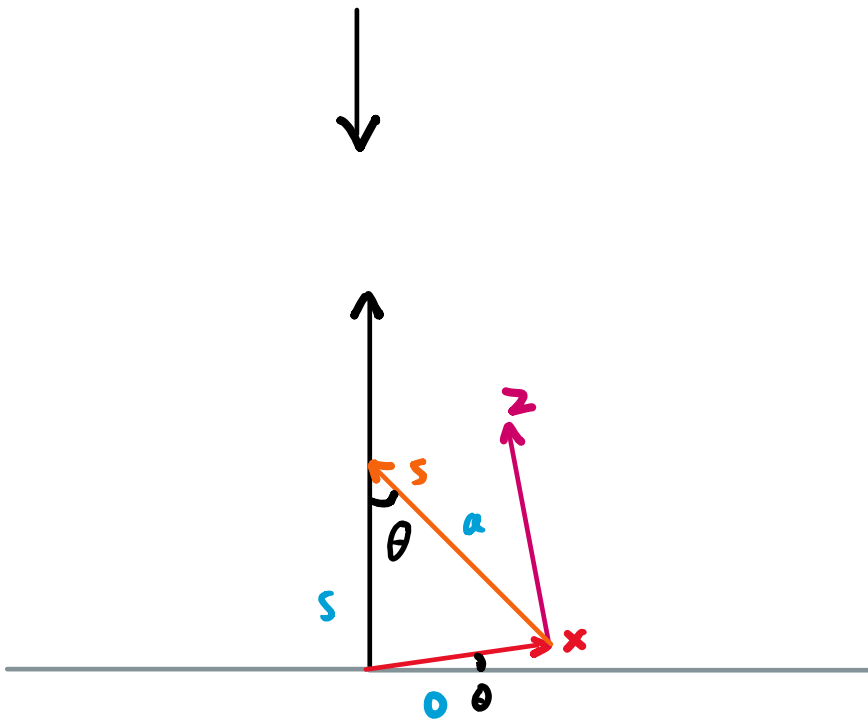


Pitch angle is $^{-1}\tan\left(\frac{\text{Accel-X}}{\text{Accel-Z}}\right) = \varphi$

• ↑ Voor Simple applications.

Roll around y-axis





• 3D gezien

$$\text{Angle } x = \tan^{-1} \left(\frac{A_y}{\sqrt{A_x^2 + A_z^2}} \right)$$

$$\text{Angle } y = \tan^{-1} \left(\frac{A_x}{\sqrt{A_y^2 + A_z^2}} \right)$$

Nadelen

- Ruis en trillingen
- fout tijdens beweging

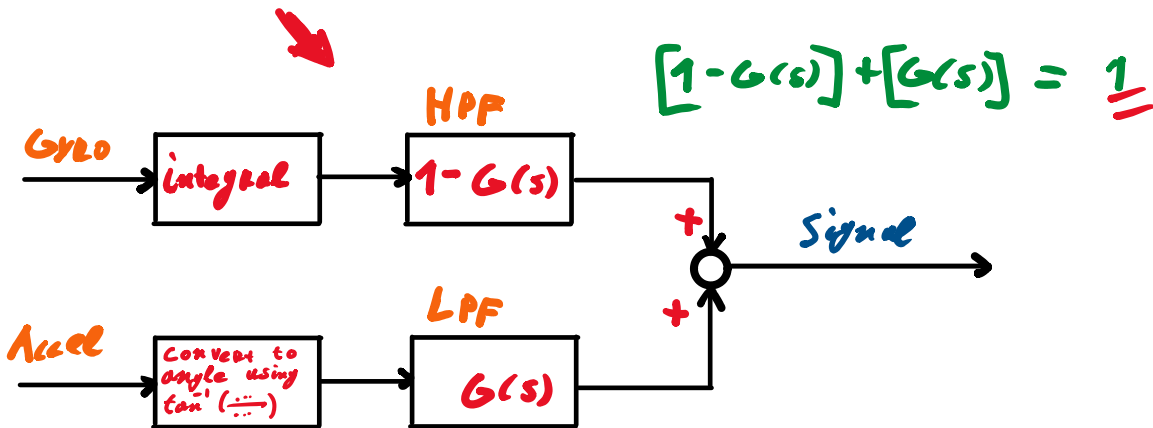




Complementary filter

⇒ Two measurements complete each other

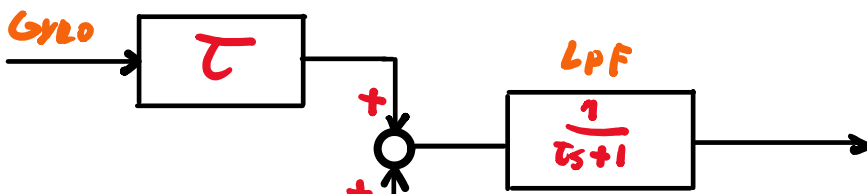
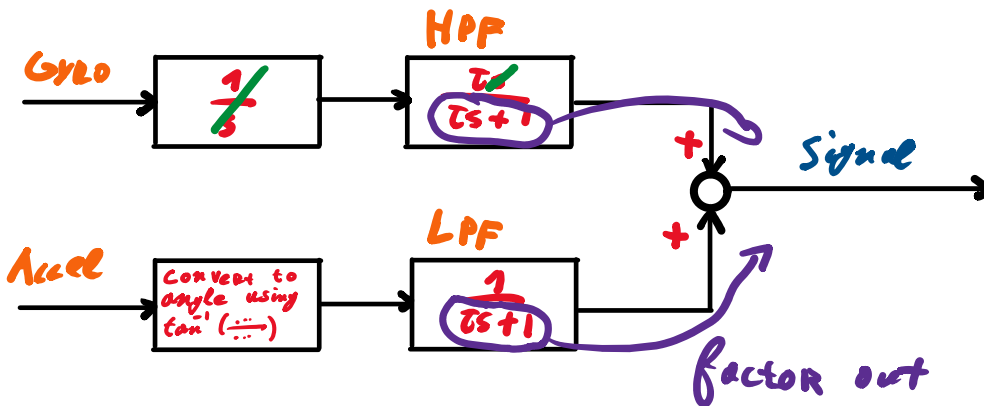
Continuous

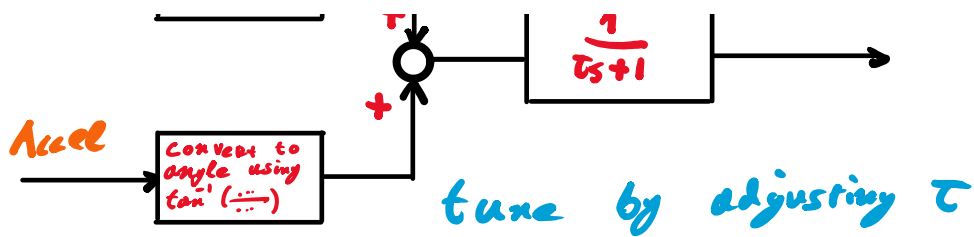


$G(s) = \frac{1}{\tau s + 1}$ ← 1st order low pass filter

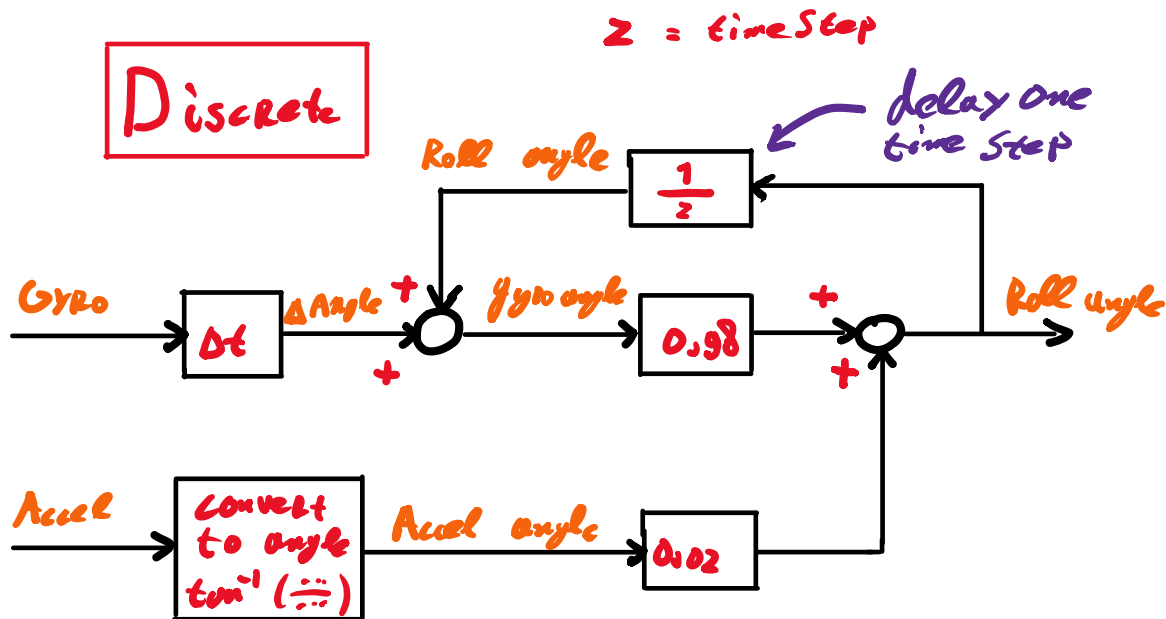
$$1 - G(s) = 1 - \frac{1}{\tau s + 1} = \frac{\tau s}{\tau s + 1}$$

integral = $\frac{1}{s}$





Low τ = Raises cutoff and more accel
 High τ = Lower cutoff and less accel



- 98% Gyro
- 2% Accel

$$\text{Roll angle} = \{ \text{Accel angle} \cdot 0.02 \} + \left\{ 0.98 \cdot \left[\text{Gyro} \cdot \Delta t + \frac{\text{Roll angle}}{z} \right] \right\}$$